
ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Department of Information Technology

Synopsis

on

IRRIGO : Precision Watering for a Sustainable Future.

1. INTRODUCTION

1.1 Overview

Agriculture has always been a critical sector for global food security, yet it remains highly dependent on natural climatic conditions such as rainfall and temperature. In developing countries like India, water scarcity and inefficient irrigation practices often result in poor crop yields and wastage of precious resources. Traditional irrigation methods require frequent human intervention, which is time-consuming and often inaccurate due to the lack of real-time data.

The project *Irrigo* aims to solve these issues by integrating **Internet of Things (IoT)** and **Machine Learning (ML)** technologies to create an intelligent, automated irrigation system. It collects real-time environmental parameters such as soil moisture, temperature, and humidity using sensors interfaced with a NodeMCU or ESP32 microcontroller. This data is transmitted to the **Blynk Cloud platform**, allowing farmers or users to monitor conditions remotely via a smartphone or web interface.

Furthermore, the collected data is stored in a database such as **MongoDB** or **Firebase**, where it can be analyzed using ML algorithms to predict optimal irrigation schedules. The system can automatically activate or deactivate water pumps based on soil conditions and weather trends, ensuring efficient water usage and improved agricultural productivity. Hence, *Irrigo* bridges the gap between

traditional farming and digital agriculture through intelligent automation and data analytics.

1.2 Purpose

The main purpose of *Irrigo* is to optimize the irrigation process by enabling **smart decision-making** using real-time sensor data and predictive analytics. The project not only provides automation but also empowers farmers with **data-driven insights** to plan and manage irrigation activities effectively.

The specific objectives include:

- **Efficient Water Management:** By accurately monitoring soil moisture and climatic conditions, the system ensures that water is supplied only when necessary, thereby reducing wastage.
- **Automation:** The system eliminates the need for manual switching of pumps by automating irrigation based on environmental conditions.
- **Predictive Intelligence:** Through the integration of machine learning algorithms, the system can analyze historical data to predict future irrigation requirements and patterns.
- **Remote Monitoring:** The Blynk mobile application allows users to view real-time data, receive notifications, and control irrigation from anywhere.
- **Scalability:** The project design can be scaled from small gardens to large agricultural fields with minimal modification.
- **Sustainability:** Promotes sustainable farming practices by reducing excessive water use and improving crop health.

By fulfilling these objectives, *Irrigo* contributes to smart agriculture, which is a step toward achieving sustainable food production and water conservation.

2. LITERATURE SURVEY

2.1 Existing Problem

Water management in agriculture is one of the major challenges in modern farming. Traditional irrigation methods like flood irrigation and manual watering are inefficient because they do not consider the actual moisture needs of the soil. As a result, these practices lead to over-irrigation or under-irrigation, which can harm plant health and reduce crop yield.

Although recent developments in IoT have led to automated irrigation systems, many existing solutions have limitations:

- They provide only **monitoring** capabilities without intelligent **decision-making**.
- Some systems depend entirely on user input, lacking adaptability to changing weather or soil conditions.
- Few systems utilize the potential of **cloud computing and machine learning** for predictive irrigation.
- Manual data recording and lack of integration with mobile applications make them less user-friendly.
- Limited accessibility in rural areas due to complex setups or lack of affordable components.

In short, current irrigation solutions either focus on automation without intelligence or analytics without real-time control. Hence, there is a need for a hybrid approach that combines IoT-based automation with data analytics and machine learning.

2.2 Proposed Solution

The proposed system, *Irrigo*, overcomes the limitations of existing systems by combining IoT, cloud computing, and machine learning technologies into a unified framework.

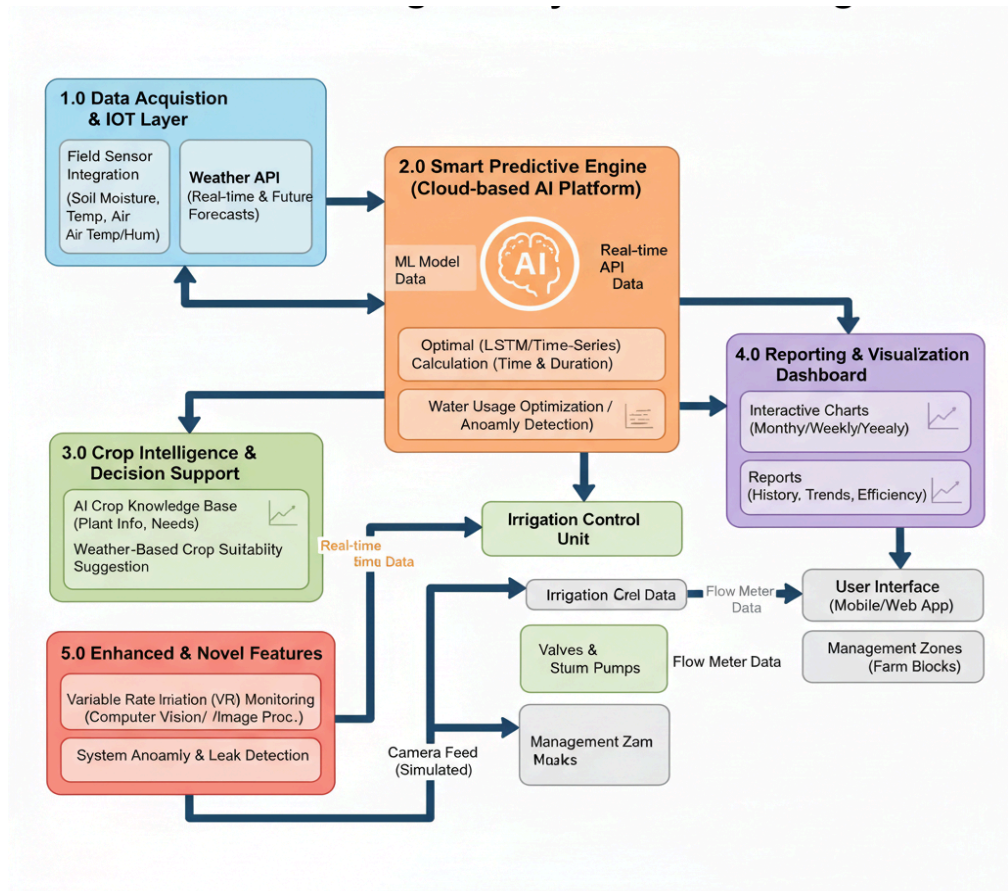
The system works as follows:

1. **Data Acquisition:** Sensors (Soil Moisture and DHT11/DHT22) measure real-time soil and environmental parameters. These sensors are connected to an ESP32 or NodeMCU microcontroller that gathers and processes the data.
2. **Data Transmission:** The sensor data is transmitted wirelessly to the Blynk Cloud platform and simultaneously stored in a cloud database such as MongoDB.
3. **Monitoring:** The Blynk mobile application provides live visualization of soil moisture, humidity, and temperature. It also allows manual control of irrigation pumps.
4. **Machine Learning Analysis:** Historical sensor data is processed using machine learning algorithms (e.g., Linear Regression or Decision Trees) to predict soil moisture trends and future irrigation needs.
5. **Automation and Control:** Based on threshold values and ML predictions, the system automatically activates or deactivates a relay module that controls the water pump.
6. **User Alerts and Reports:** Users receive alerts about abnormal conditions, and detailed reports can be viewed through the dashboard.

This proposed solution ensures optimized water use, minimal human intervention, and increased productivity through smart prediction and control.

3. THEORETICAL ANALYSIS

3.1 Block Diagram



Description of System Components:

1. Sensors:

- **Soil Moisture Sensor:** Detects the water content in the soil. When the moisture level falls below a threshold, it signals the controller to initiate irrigation.
- **DHT11/DHT22 Sensor:** Measures ambient temperature and humidity, essential for analyzing evaporation rates and crop health.

2. Microcontroller (ESP32/NodeMCU):

Acts as the central processing unit. It collects data from sensors, processes it, and communicates with the cloud via Wi-Fi.

3. **Wi-Fi Module:**

Enables wireless communication between the microcontroller and the Blynk Cloud.

4. **Cloud Platform (Blynk Cloud):**

Provides a real-time interface for data visualization and remote control through a smartphone.

5. **Database (MongoDB/Firebase):**

Stores historical data for further analytics and ML training.

6. **Machine Learning Model:**

Uses collected data to predict optimal irrigation times and quantities, reducing unnecessary watering.

7. **Relay and Pump:**

The relay module acts as an electronic switch to control the water pump based on the system's decisions.

8. **Mobile Application:**

Displays live readings, alerts, and allows remote control for manual override

3.2 Hardware/Software Designing

3.2.1 Hardware Requirements:

The project requires a set of sensors and microcontrollers to collect environmental data and control irrigation. The main hardware components include:

- Arduino Uno – Acts as the central microcontroller for processing sensor data and controlling outputs.
- Soil Moisture Sensor – Measures the water content in the soil to determine irrigation needs.
- DHT11 Sensor – Monitors temperature and humidity levels of the environment.
- Water Pump Module – Automatically turns ON or OFF based on soil moisture readings.

- Relay Module – Used to control the water pump operation safely.
- Power Supply (5V/12V) – Provides power to the Arduino and other modules.
- Wi-Fi Module (ESP8266 or NodeMCU) – Sends sensor data to the Blynk Cloud for real-time monitoring.

3.2.2 Software Requirements:

The project requires a set of sensors and microcontrollers to collect environmental data and control irrigation. The main hardware components include:

- **Arduino Uno** – Acts as the central microcontroller for processing sensor data and controlling outputs.
- **Soil Moisture Sensor** – Measures the water content in the soil to determine irrigation needs.
- **DHT11 Sensor** – Monitors temperature and humidity levels of the environment.
- **Water Pump Module** – Automatically turns ON or OFF based on soil moisture readings.
- **Relay Module** – Used to control the water pump operation safely.
- **Power Supply (5V/12V)** – Provides power to the Arduino and other modules.
- **Wi-Fi Module (ESP8266 or NodeMCU)** – Sends sensor data to the Blynk Cloud for real-time monitoring.

4. APPLICATIONS

The *Irrigo* system has wide-ranging applications in both agricultural and non-agricultural sectors:

1. **Smart Agriculture:** Automates irrigation for large-scale farming operations, improving efficiency and yield.

2. **Greenhouse Monitoring:** Maintains ideal conditions for crops by monitoring humidity and temperature.
3. **Urban Gardening:** Enables home gardeners to maintain plants without manual watering.
4. **Government and NGOs:** Useful in smart village projects and sustainable farming programs.
5. **Research Institutions:** Can be used for agricultural data collection and model validation.
6. **Water Resource Management:** Helps in efficient water distribution in drought-prone or water-scarce regions.

REFERENCES

- [1] Government of India, “*Open Government Data Platform (data.gov.in)*” — *Agriculture Crop & Weather Datasets*.
- [2] Hughes, David, *Smart Agriculture — AI & IoT-Enabled Farming*, Springer, 2022.
- [3] Jha, A., Neeraj, D., and Garg, R., “*Smart Irrigation System Using IoT and Machine Learning*,” IEEE Conference on Emerging Devices and Smart Systems (ICEDSS), 2021.
- [4] Kim, S., Kim, M., and Kim, D., “*Deep Learning-Based Crop Model for Smart Farming*,” IEEE Access, vol. 8, pp. 181535–181545, 2020.
- [5] NASA, “*POWER Climate API — Solar Radiation, Humidity, and Temperature Datasets for Agriculture*.”

Guided By:

Prof. Mahendra Verma

Group Members:

Om Patil (0827IT221102)

Riddhi Pathak (0827IT221120)

Saloni Jain (0827IT221128)

Sarthak Kacholiya (0827IT221132)